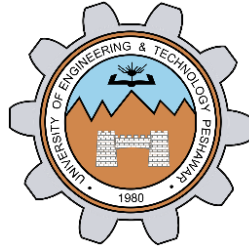


Interfacing an LCD to the 8051 Microcontroller

LAB # 10



Spring 2024

CSE-307L Microprocessor Based System Design Lab

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Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Dr. Asif Ali Khan

Date:

23rd May 2024

Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar

Task 1:

Write Your Name on First Line and registration Number on second line of LCD.

Code:

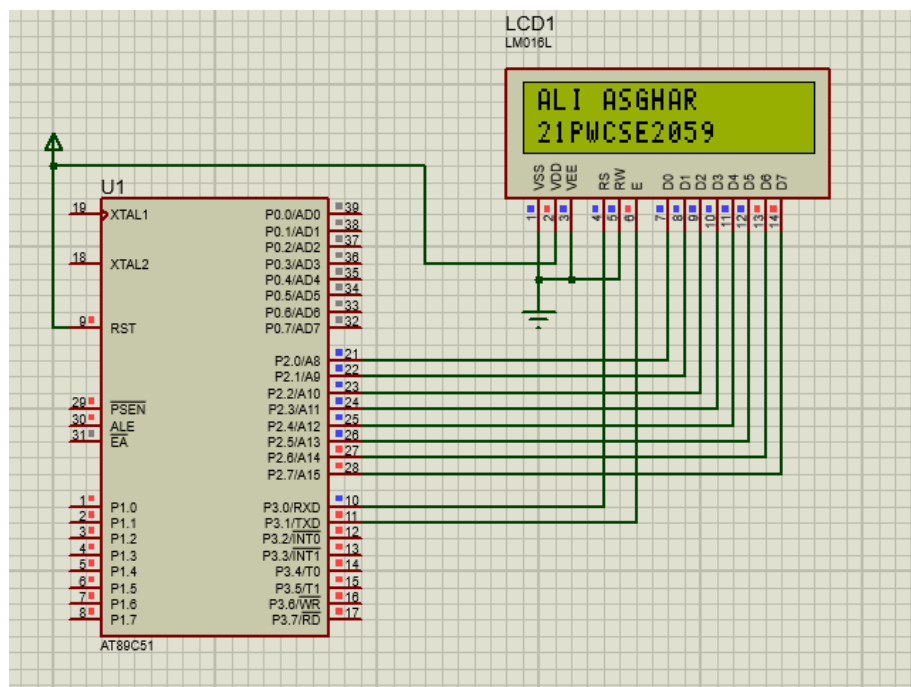
```
lab_task.c | STARTUP.A51
1  #include<reg51.h>
2  //RS = 1 we are passing data
3
4  sbit RS = P3^0;
5  sbit Enable = P3^1;
6
7  void init();
8  void delay();
9  void wr_cmd(unsigned char);
10 void wr_data(unsigned char);
11
12 char *name = "ALI ASGHAR";
13 char *reg_no = "21PWCSE2059";
14
15 void main(void){
16     P2 = 0x00;
17     init();
18
19     while(1){
20         while(*name)
21             wr_data(*name++);
22
23         wr_cmd(0xc0); //2nd line selected
24
25         while(*reg_no)
26             wr_data(*reg_no++);
27     }
28 }
29
30
31 void init(){
32     wr_cmd(0x38); //2 lines and 5x7 Matrix
33     wr_cmd(0x0e); //Display on and cursor blinking
34     wr_cmd(0x01); //Clear Display Screen
35     wr_cmd(0x06); //Increment Cursor
36     wr_cmd(0x0c); //Display on, Cursor off
37     wr_cmd(0x80); //Force cursor to beginning of 1st l
38 }
39
40
41 void wr_cmd(unsigned char cmd){
42     P2 = cmd;
43     RS = 0;
44     Enable = 1;
45     delay();
46     Enable = 0;
47 }
48
49 void wr_data(unsigned char dt){
```

```

48 |
49 | void wr_data(unsigned char dt){
50 |     P2 = dt;
51 |     RS = 1;
52 |     Enable = 1;
53 |     delay();
54 |     Enable = 0;
55 |
56 | }
57 |
58 | void delay(){
59 |     unsigned int i;
60 |
61 |     for(i = 0; i<15000; i++);
62 | }
63 |
64 |

```

Proteus Simulation:



Task 2:

Input numbers from keypad from 1 to 9 and display on the LCD

Code:

```
lab_task2.c  STARTUP.A51
1  #include<reg51.h>
2  //RS = 1 we are passing data
3
4  sbit RS = P3^0;
5  sbit Enable = P3^1;
6  sbit R1 = P1^0;
7  sbit R2 = P1^1;
8  sbit R3 = P1^2;
9  sbit R4 = P1^3;
10
11 sbit C1 = P1^4;
12 sbit C2 = P1^5;
13 sbit C3 = P1^6;
14
15 void init();
16 void delay(unsigned int);
17 void wr_cmd(unsigned char);
18 void wr_data(unsigned char);
19 void show(unsigned char);
20 unsigned char key_scanning();
21 int get_num(char);
22 void disp_num(int);
23
lab_task2.c  STARTUP.A51
22 void disp_num(int);
23
24 unsigned char num_char;
25 int num;
26
27 void main(void){
28     P2 = 0x00;
29     init();
30
31     while(1){
32         num_char = key_scanning();
33         num = get_num(num_char);
34         disp_num(num);
35     }
36 }
37
38
39
40 void init(){
41     wr_cmd(0x38); //2 lines and 5x7 Matrix
42     wr_cmd(0x0e); //Display on and cursor blinking
43     wr_cmd(0x01); //Clear Display Screen
44     wr_cmd(0x06); //Increment Cursor
```

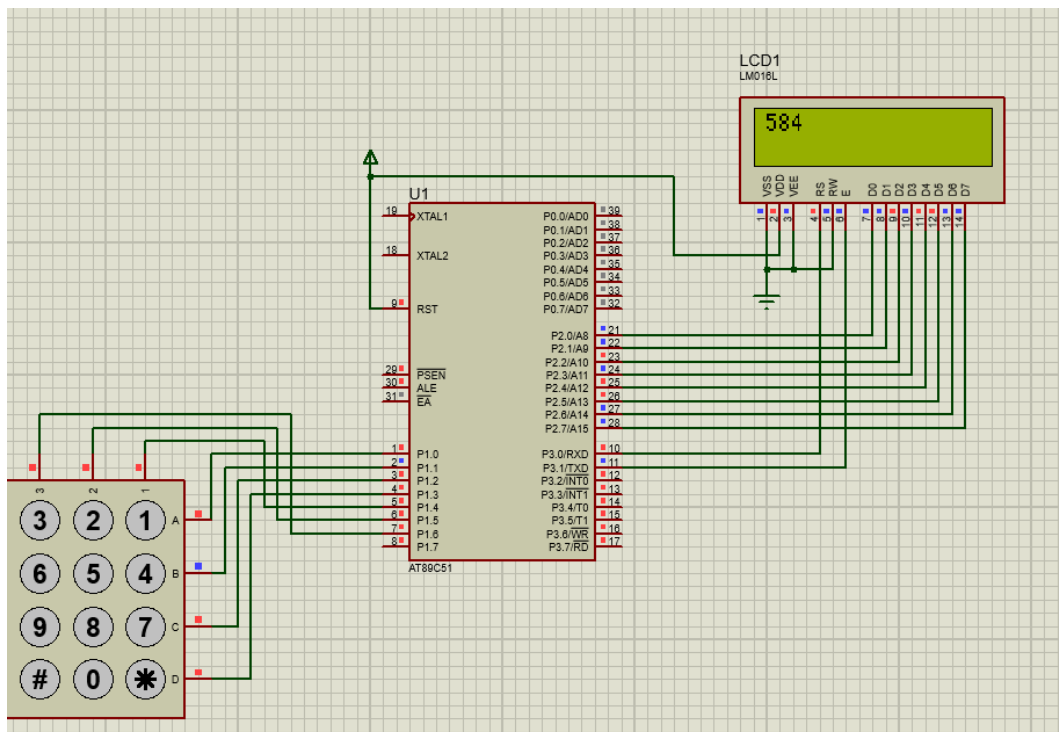
```
lab_task2.c  STARTUP.A51
43 wr_cmd(0x01); //Clear Display Screen
44 wr_cmd(0x06); //Increment Cursor
45 wr_cmd(0x0c); //Display on, Cursor off
46 wr_cmd(0x80); //Force cursor to begining of 1st line
47
48 }
49
50 void wr_cmd(unsigned char cmd) {
51     P2 = cmd;
52     RS = 0;
53     Enable = 1;
54     delay(15000);
55     Enable = 0;
56 }
57
58 void wr_data(unsigned char dt) {
59     P2 = dt;
60     RS = 1;
61     Enable = 1;
62     delay(15000);
63     Enable = 0;
64 }
65
```

```
lab_task2.c  STARTUP.A51
64 }
65
66 void delay(unsigned n) {
67     unsigned int i;
68
69     for(i = 0; i<n; i++);
70 }
71
72 unsigned char key_scanning() {
73
74     while(1) {
75         R1 = 0;
76         R2 = R3 = R4 = 1;
77         if(C1 == 0 && C2 == 1 && C3 == 1) { //1 Pressed
78             delay(10000);
79             return '1';
80         }
81
82         if(C1 == 1 && C2 == 0 && C3 == 1) { //2 Pressed
83             delay(10000);
84             return '2';
85         }
86     }
87 }
```

```
lab_task2.c  STARTUP.A51
86
87     if(C1 == 1 && C2 == 1 && C3 == 0){//3 Pressed
88         delay(10000);
89         return '3';
90     }
91
92
93     R2 = 0;
94     R1 = R3 = R4 = 1;
95     if(C1 == 0 && C2 == 1 && C3 == 1){//4 Pressed
96         delay(10000);
97         return '4';
98     }
99
100    if(C1 == 1 && C2 == 0 && C3 == 1){//5 Pressed
101        delay(10000);
102        return '5';
103    }
104
105    if(C1 == 1 && C2 == 1 && C3 == 0){//6 Pressed
106        delay(10000);
107        return '6';
108    }
```

```
lab_task2.c  STARTUP.A51
107     return '6';
108 }
109
110
111     R3 = 0;
112     R1 = R2 = R4 = 1;
113
114     if(C1 == 0 && C2 == 1 && C3 == 1){//7 Pressed
115         delay(10000);
116         return '7';
117     }
118
119     if(C1 == 1 && C2 == 0 && C3 == 1){//8 Pressed
120         delay(10000);
121         return '8';
122     }
123
124     if(C1 == 1 && C2 == 1 && C3 == 0){//9 Pressed
125         delay(10000);
126         return '9';
127     }
128
129     R4 = 0;
```

Proteus Simulation:



Task 3:

Make a calculator using keypad and LCD which perform (+,-,*,/)

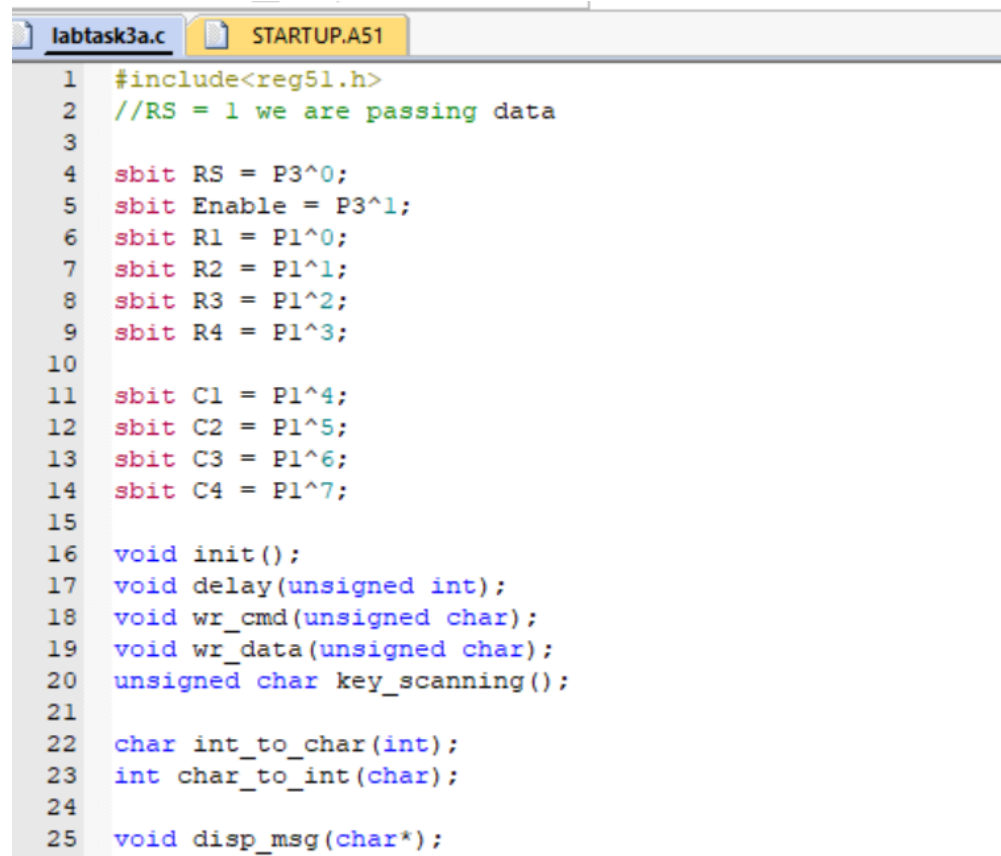
$$6 + 6 = 12$$

$$6 * 6 = 36$$

$$6 / 6 = 1$$

$$6 - 3 = 3$$

Code:



```
1  #include<reg51.h>
2  //RS = 1 we are passing data
3
4  sbit RS = P3^0;
5  sbit Enable = P3^1;
6  sbit R1 = P1^0;
7  sbit R2 = P1^1;
8  sbit R3 = P1^2;
9  sbit R4 = P1^3;
10
11 sbit C1 = P1^4;
12 sbit C2 = P1^5;
13 sbit C3 = P1^6;
14 sbit C4 = P1^7;
15
16 void init();
17 void delay(unsigned int);
18 void wr_cmd(unsigned char);
19 void wr_data(unsigned char);
20 unsigned char key_scanning();
21
22 char int_to_char(int);
23 int char_to_int(char);
24
25 void disp_msg(char*);
```



```

labtask3a.c  STARTUP.A51
74     wr_cmd(0x06); //Increment Cursor
75     wr_cmd(0x0c); //Display on, Cursor off
76     wr_cmd(0x80); //Force cursor to beginning of 1st lin
77
78 }
79
80 void wr_cmd(unsigned char cmd){
81     P2 = cmd;
82     RS = 0;
83     Enable = 1;
84     delay(10000);
85     Enable = 0;
86 }
87
88 void wr_data(unsigned char dt){
89     P2 = dt;
90     RS = 1;
91     Enable = 1;
92     delay(10000);
93     Enable = 0;
94 }
95 }
96
97 void delay(unsigned n){
98     unsigned int i;

```

```

labtask3a.c  STARTUP.A51
98     unsigned int i;
99
100     for(i = 0; i<n; i++){
101     }
102
103 unsigned char key_scanning(){
104
105 while(1){
106     R1 = 0;
107     R2 = R3 = R4 = 1;
108     if(C1 == 0 && C2 == 1 && C3 == 1 && C4 == 1){//7 Pressed
109         delay(10000);
110         return '7';
111     }
112
113     if(C1 == 1 && C2 == 0 && C3 == 1 && C4 == 1){//8 Pressed
114         delay(10000);
115         return '8';
116     }
117
118     if(C1 == 1 && C2 == 1 && C3 == 0 && C4 == 1){//9 Pressed
119         delay(10000);
120         return '9';
121     }
122

```

```
labtask3a.c  STARTUP.A51
123  if(C1 == 1 && C2 == 1 && C3 == 1 && C4 == 0){///<% Pressed
124      delay(10000);
125      return '/';
126  }
127
128  R2 = 0;
129  R1 = R3 = R4 = 1;
130  if(C1 == 0 && C2 == 1 && C3 == 1 && C4 == 1){///<4 Pressed
131      delay(10000);
132      return '4';
133  }
134
135  if(C1 == 1 && C2 == 0 && C3 == 1 && C4 == 1){///<5 Pressed
136      delay(10000);
137      return '5';
138  }
139
140  if(C1 == 1 && C2 == 1 && C3 == 0 && C4 == 1){///<6 Pressed
141      delay(10000);
142      return '6';
143  }
144
145  if(C1 == 1 && C2 == 1 && C3 == 1 && C4 == 0){///
```

```
labtask3a.c  STARTUP.A51
147      return 'x';
148  }
149
150  R3 = 0;
151  R1 = R2 = R4 = 1;
152
153  if(C1 == 0 && C2 == 1 && C3 == 1 && C4 == 1){///<1 Pressed
154      delay(10000);
155      return '1';
156  }
157
158  if(C1 == 1 && C2 == 0 && C3 == 1 && C4 == 1){///<2 Pressed
159      delay(10000);
160      return '2';
161  }
162
163  if(C1 == 1 && C2 == 1 && C3 == 0 && C4 == 1){///<3 Pressed
164      delay(10000);
165      return '3';
166  }
167
168  if(C1 == 1 && C2 == 1 && C3 == 1 && C4 == 0){///<- Pressed
169      delay(10000);
170      return '-';
171  }
```

```
labtask3a.c  STARTUP.A51
174     R1 = R2 = R3 = 1;
175
176     if(C1 == 0 && C2 == 1 && C3 == 1 && C4 == 1){//C/ON Pressed
177         delay(10000);
178         return 'C';
179     }
180
181     if(C1 == 1 && C2 == 0 && C3 == 1 && C4 == 1){//0 Pressed
182         delay(10000);
183         return '0';
184     }
185
186     if(C1 == 1 && C2 == 1 && C3 == 0 && C4 == 1){//= Pressed
187         delay(10000);
188         return '=';
189     }
190
191     if(C1 == 1 && C2 == 1 && C3 == 1 && C4 == 0){//+ Pressed
192         delay(10000);
193         return '+';
194     }
195 }
196 }
197
198 char int_to_char(int num){
```

```
labtask3a.c  STARTUP.A51
198 char int_to_char(int num){
199     return num + '0';
200 }
201
202
203 int char_to_int(char num){
204     return num - '0';
205 }
206
207 void disp_msg(char *str){
208     while(*str)
209         wr_data(*str++);
210 }
211
212 void get_result(){
213
214     switch(operation){
215         case '+': res_num = num1 + num2; break;
216         case '-': res_num = num1 - num2; break;
217         case 'x': res_num = num1 * num2; break;
218         case '/': res_num = num1 / num2; break;
219         default:
220             disp_msg("Invalid");
221             wr_cmd(0x01); //Clear Display Screen
222             main();
```

```

225     res_buff_int[0] = res_num / 10;
226     res_num = res_num % 10;
227     res_buff_int[1] = res_num;
228
229 }
230 void display_result() {
231     //wr_cmd(0x01); //Clear Display Screen
232     wr_cmd(0xC0); //Move to 2nd line
233     disp_msg(buff);
234     disp_msg(" = ");
235     wr_data(int_to_char(res_buff_int[0]));
236     wr_data(int_to_char(res_buff_int[1]));
237     wr_cmd(0x80); //Move to 1st line
238 }
239

```

Proteus Simulation:

